TEXORA · LECTURE SHEET

# Fiber Migration

*A fibre in a spun yarn does not stay at one depth. As the yarn runs on, each fibre spirals in towards the core and out towards the surface again — and that movement, fibre migration, quietly decides yarn strength, hairiness, handle and dyeing appearance. This sheet covers what migration is, why it happens, how it is described, and where each fibre of a blend prefers to live.*

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 11 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

## 1. Definition

**Fibre migration** is the change in the radial position of a fibre along the length of a staple-fibre yarn: a fibre that lies near the surface at one cross-section is found nearer the core a short distance further along, and later returns towards the surface, following a roughly helical path whose radius continually varies. Migration is a property of staple (spun) yarns; in a filament yarn the filaments keep an essentially fixed radial position.

Migration is not a defect. It is a necessary consequence of how twist is inserted into a strand of short fibres, and a yarn with no migration at all would fall apart: it is the interlocking produced by fibres continually changing depth that binds short fibres into a coherent yarn. The practical questions are how much migration there is, whether it is regular, and — in blends — which fibres finish up where.

## 2. Radial migration — the picture

Figure 1 shows the same small group of fibres followed through four successive cross-sections of one yarn. Fibre A starts at the surface and moves steadily inward; fibre B starts near the core and moves outward; fibre C stays in the middle zone. No fibre keeps its starting radius. Over a longer length each fibre repeats this inward–outward journey many times, so a single fibre contributes alternately to the yarn surface (where it affects handle, hairiness and dye uptake) and to the core (where it is gripped most firmly and contributes most reliably to strength).

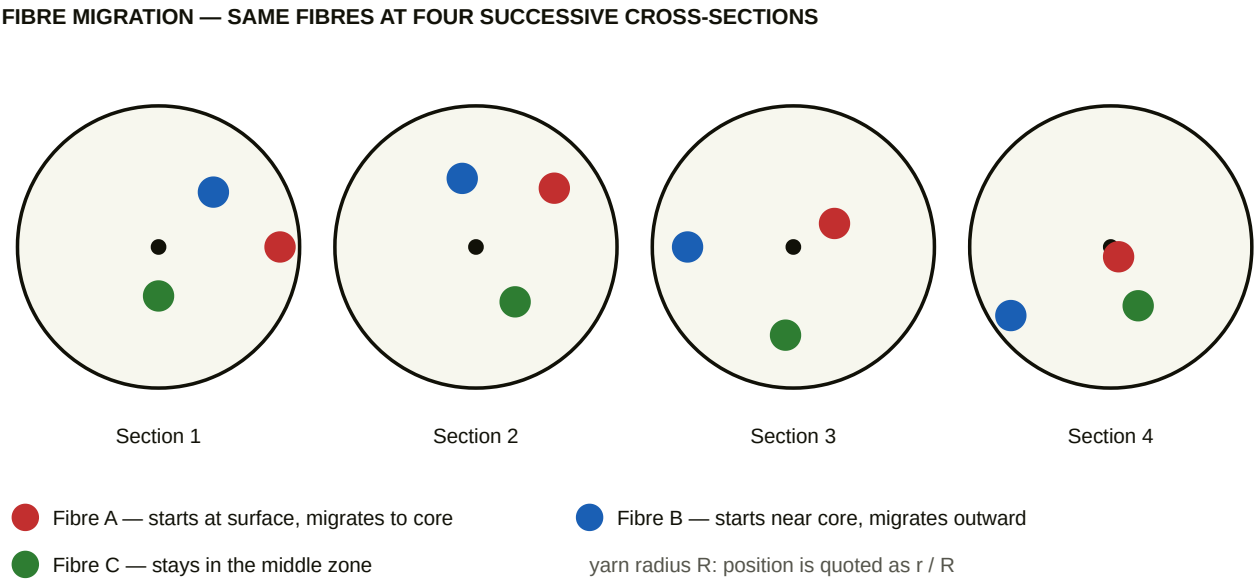



Figure 1 — Radial migration. Three fibres traced through four successive yarn cross-sections: each changes its radial position along the yarn; none stays at a fixed depth.

# 3. Causes of migration

## 3.1 Tension differences during twisting

The fundamental cause, established in the classical work of Morton and co-workers, is the difference in fibre tension during twist insertion. When twist runs into the strand at spinning, fibres do not all carry the same tension: a fibre lying on a longer, more helical path is stretched more and carries higher tension than one lying straighter. A highly tensioned fibre tends to shorten its path by moving towards the yarn axis — displacing a slacker fibre outward. As the yarn moves forward and new fibre is continually fed in, every fibre repeatedly gains and loses tension, so it repeatedly trades places with its neighbours. Migration is the visible result of this continual exchange.

## 3.2 Fibre properties

- **Length.** Longer fibres migrate over longer cycles and are gripped over more of their length; very short fibres migrate poorly, are easily lost to the surface and contribute most of the fly and weak places.
- **Fineness.** Finer fibres are more flexible, pack more closely and preferentially occupy the core (see Section 5); coarser fibres are pushed towards the surface.
- **Stiffness / flexural rigidity.** A stiff fibre resists the tight bending needed near the core and stays outward; a supple fibre bends easily and can take up an inner position.
- **Friction, crimp and cross-section** modify the picture: high-friction and highly crimped fibres resist slipping past neighbours, so their migration pattern is more locked-in; smooth round fibres exchange

places more freely.

### 3.3 Yarn and process variables

Higher twist increases the tension differences that drive migration, up to the point where the yarn becomes so compact that movement is restricted. Spinning system matters: ring-spun yarns show the classic full migration pattern; rotor (open-end) yarns show a less regular structure with wrapper fibres around a relatively untwisted core; air-jet yarns have a near-parallel core bound by surface wrapper fibres — migration in the classical sense is largely absent, which is why their strength and handle differ from ring-spun yarn.

## 4. Describing migration — parameters

In Morton's classical description a fibre's path is characterised qualitatively by three ideas, still the clearest way to report migration without heavy mathematics:

- **Mean fibre position** — the average radial position of the fibres, quoted as a fraction of the yarn radius (r/R, from 0 at the axis to 1 at the surface). A value of 0.5 represents fibres distributed evenly between core and surface; below 0.5 the fibres concentrate towards the core; above 0.5 towards the surface. This is the one simple numerical measure used in the worked example below.
- **Migration amplitude** — how far, radially, a fibre travels between its innermost and outermost positions: large amplitude means fibres sweep from core to surface; small amplitude means they hover in one zone.
- **Migration frequency** — how often, per unit yarn length, a fibre completes an inward–outward cycle: high frequency means rapid exchange of positions along the yarn.

**WORKED EXAMPLE — MEAN FIBRE POSITION**

In one yarn cross-section, six fibres are found at radial positions (as fractions of the yarn radius) of 0.10, 0.25, 0.40, 0.55, 0.70 and 0.85.
Sum = 0.10 + 0.25 + 0.40 + 0.55 + 0.70 + 0.85 = 2.85.
Mean fibre position = 2.85 / 6 = **0.475**.
Interpretation: 0.475 is just below 0.5, so in this section the fibres lie very slightly towards the core of an even distribution — essentially a well-distributed section. A single section never proves a pattern: the measure is averaged over many sections along the yarn before any conclusion is drawn.

Migration is observed experimentally by tracer-fibre methods: a few dyed fibres are blended into the stock, the yarn is made transparent or sectioned, and the dyed fibres are followed along the yarn. The

method is slow, which is why migration is usually discussed through its measurable consequences (Section 5) rather than measured on every production yarn.

## 5. Effects of migration

| PROPERTY | EFFECT OF MIGRATION |
|---|---|
| Yarn strength | Regular migration binds fibres together so load is shared and fibres are gripped over much of their length — strength rises. Poor or irregular migration leaves fibres poorly gripped; they slip instead of breaking, and strength and evenness fall. |
| Hairiness | Fibre ends that migrate outward and stay at the surface form hairs. Fibres with a surface preference (coarse, stiff) raise hairiness; a core-preferring distribution lowers it. |
| Fabric hand and appearance | The fibre that occupies the yarn surface dominates fabric hand, lustre and abrasion behaviour — in a blend, the surface fibre is the one the wearer feels and sees first. |
| Dyeing appearance | If blend components dye differently and one concentrates at the surface, the fabric shows that component's shade disproportionately; uneven migration along the yarn shows as streaky, barre-like dyeing faults. |
| Pilling and abrasion | Surface fibres are abraded and pilled first; a strong fibre at the surface anchors pills, a weak fibre at the surface sheds them — migration position therefore changes pilling behaviour. |

## 6. Migration in blended yarns

In a blend, the components do not distribute randomly: each fibre type takes up a preferred radial position, following the property rules of Section 3. As a general, well-established pattern:

- **Longer fibres tend to the core; shorter fibres to the surface** — the short fibres are progressively displaced outward and are the first lost as fly.
- **Finer fibres tend to the core; coarser fibres to the surface,** because fine fibres bend and pack more easily in the centre.
- **Less stiff (more supple) fibres tend to the core; stiff fibres stay outward.**

Worked consequences follow directly. In a polyester/cotton blend where the polyester is longer and finer than the cotton, polyester concentrates towards the core and cotton towards the surface: the fabric feels like cotton, dyes (in the cotton phase) like cotton, yet draws much of its strength from the polyester core — a deliberate and useful design. Reverse the fibre dimensions and the surface character reverses with them. In wool blends, the finer component similarly takes the core. The design rule is therefore: **choose which fibre you want the customer to feel, see and dye — then give that fibre the**

**coarser, shorter, stiffer character if you want it at the surface, or blend it in knowing it will be partly hidden if it is longer and finer than its partner.** Ignoring preferential migration is a classic cause of a blend that passes its fibre-content test yet looks, feels and dyes "wrong".

Preferential migration also changes with blend ratio and with processing: it is strongest when the two components differ clearly in length, fineness or stiffness, and it can be moderated by matching those properties between the components when an even distribution is wanted.

## Glossary

**Fibre migration** — change of a fibre's radial position along a staple yarn. **Radial position (r/R)** — a fibre's distance from the yarn axis as a fraction of the yarn radius. **Mean fibre position** — average radial position of the fibres in a section or yarn. **Migration amplitude** — radial distance a fibre travels between its innermost and outermost positions. **Migration frequency** — number of inward–outward migration cycles per unit yarn length. **Preferential migration** — tendency of one blend component to occupy the core or the surface. **Tracer fibre** — a dyed fibre blended in small quantity so its path can be followed. **Wrapper fibre** — a fibre wound around the surface of a rotor- or air-jet-spun yarn, binding its core.

## Key points

• Migration = fibres continually changing radial position along the yarn; it binds staple fibres into a coherent yarn.
• Cause: tension differences during twisting — a highly tensioned fibre cuts in towards the axis and displaces a slacker fibre outward.
• Longer, finer and more supple fibres prefer the core; shorter, coarser and stiffer fibres are displaced to the surface.
• Described by mean fibre position (r/R average), migration amplitude and migration frequency (Morton).
• Regular migration raises strength; surface distribution decides hairiness, hand, dyeing appearance and pilling.
• In blends, preferential migration decides which fibre the fabric shows — design for it, do not discover it after dyeing.

## Practice

1. Five fibres in a yarn cross-section lie at radial positions 0.0, 0.2, 0.5, 0.8 and 1.0 (fractions of the yarn radius). Calculate the mean fibre position and state whether the fibres in this section concentrate towards the core, the surface, or neither.

2. A polyester/cotton yarn is spun from polyester that is longer and finer than the cotton. Predict which fibre occupies the surface, and state two fabric consequences.
3. Explain why an air-jet yarn, whose core fibres are nearly parallel and bound by wrapper fibres, behaves differently in strength and handle from a ring-spun yarn of the same count and blend. Refer to migration in your answer.
4. A blended fabric dyes streakily although the blend ratio tests correctly in every sample. Give the migration-based explanation and one processing change that would reduce the fault.

## Answer key

1. Sum = 0.0 + 0.2 + 0.5 + 0.8 + 1.0 = 2.5. Mean = 2.5 / 5 = **0.50**. Neither: 0.50 is exactly the even-distribution value, so in this section the fibres are spread evenly between core and surface.
2. The **cotton** occupies the surface: it is the shorter, coarser component, while the longer, finer polyester migrates to the core. Consequences (any two): the fabric hand and appearance are cotton-like; dyeing is dominated by the cotton shade at the surface; strength is supported by the polyester core; the cotton surface abrades and pills as cotton does.
3. In the ring-spun yarn, migration continually interlocks every fibre between core and surface, so fibres are gripped along their length, share load and give a full, fibrous handle. In the air-jet yarn the core fibres barely migrate — they lie parallel and are held only by the surface wrappers — so fibres slip more easily (lower strength utilisation) and the yarn is leaner and crisper in handle, with properties set by the wrappers rather than by an interlocked structure.
4. The blend components are migrating unevenly along the yarn (preferential migration varying from place to place), so the surface — which is what dyeing shows — varies in composition even though the bulk blend ratio is constant. Reduce it by matching the components' length, fineness and stiffness more closely (or by improving blending/drafting regularity), so neither component preferentially and variably takes the surface.

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